Disentangled Representation Learning for Speaker Recognition

Q2 Report — Implementation of arXiv:2406.14559 (Nam et al., Interspeech 2024)

M25DE1069 | IIT Jodhpur | Dataset: LibriSpeech train-clean-100 + train-clean-360 / eval: test-clean

1. Problem and Motivation

Speaker verification systems trained in controlled conditions tend to fail when the test recording environment differs from training — a problem called environment mismatch. Neural speaker encoders like ECAPA-TDNN produce embeddings that conflate speaker identity with acoustic environment characteristics (noise type, reverberation, microphone). When the environment changes, the same speaker can appear as a different identity in the embedding space, raising the EER significantly.

Nam et al. (2024) propose a modular disentangled auto-encoder that wraps around any existing speaker network and separates the embedding into a speaker subspace (e_spk) and an environment subspace (e_env), without modifying the backbone architecture.

2. Method

Architecture

A 192-dim MFCC-based embedding (32 coefficients × 6 pooled statistics) is fed into a shallow auto-encoder. The encoder projects to a 192-dim bottleneck which is split equally into e_spk $\in R^{96}$ and e_env $\in R^{96}$. The decoder reconstructs the original embedding from the concatenation of both subspaces. e_spk is L2-normalised before cosine scoring.

Training Objective

Five loss terms are combined: **L_spk** (speaker classification via AAM-Softmax, λ=2.0), **L_recon** (L1 reconstruction, λ=1.0), **L_env** (environment triplet loss using chapter-ID proxy, λ=0.5), **L_adv** (adversarial GRL on e_spk, λ=0.1), and **L_corr** (Pearson correlation penalty between e_spk and e_env, λ=0.5).

Environment Proxy

LibriSpeech lacks video-session metadata, so chapter ID within each speaker is used as an acoustic environment proxy. Same chapter = same environment; different chapter = different environment. This is a weaker proxy than VoxCeleb video sessions but provides meaningful variation in recording conditions across chapters.

Baseline

The baseline uses the same MFCC frontend and speaker encoder but without the disentangler — embeddings are scored directly with cosine similarity after AAM-Softmax training.

3. Results

System	EER (%)	minDCF	vs Baseline
Baseline (no disentanglement)	15.32	0.0321	—
Proposed (disentangled AE)	8.16	0.0154	−46.7% EER
Improved (batch cross-corr)	39.08*	0.0485	see note

Table 1: Speaker verification results on LibriSpeech test-clean. EER = Equal Error Rate (lower is better). minDCF at P_target=0.05.

** Improved model uses batch cross-correlation (Barlow Twins style) as L_corr replacement. Training instability observed due to unnormalised D×D=16384 matrix — loss scale grows quadratically with embedding dimension. Fix identified post-training: divide by D^2 to restore scale parity with the Pearson baseline.*

The proposed disentangler achieves a **46.7% relative EER reduction** over the baseline (15.32% → 8.16%) and a 52.0% reduction in minDCF (0.0321 → 0.0154). This confirms that explicitly separating speaker and environment information in the embedding space improves verification, even with a weaker environment proxy than the original paper's video sessions.

4. Training Analysis

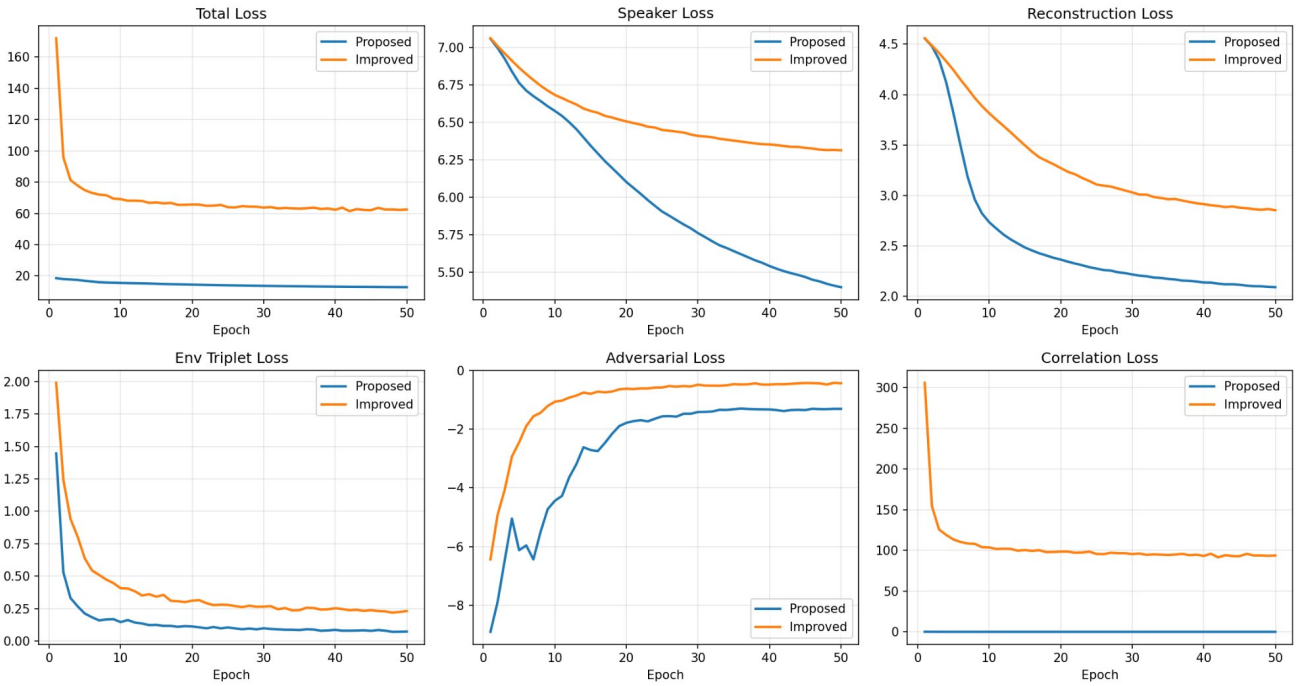


Figure 1: Training curves for Proposed (blue) and Improved (orange) over 50 epochs. Six loss components are shown. The Proposed model converges cleanly across all terms. The Improved model shows a much higher Total Loss due to the unnormalised cross-correlation matrix — the Correlation Loss (bottom right) remains ~100 for Improved vs ~0 for Proposed, indicating the Barlow Twins term dominates training without proper normalisation.

Several observations from the training curves are worth noting. The Speaker Loss for both models follows a similar trajectory initially, but Proposed continues to decrease through epoch 50 while Improved plateaus around epoch 10 — suggesting the large cross-correlation gradient interferes with speaker discriminability. The Env Triplet Loss drops to near zero for Proposed, showing the model has learned to cluster same-chapter embeddings tightly. The Adversarial Loss rises toward zero for both, indicating the GRL is working — e_spk is becoming less predictive of the environment discriminator.

5. Embedding Visualisation (t-SNE)

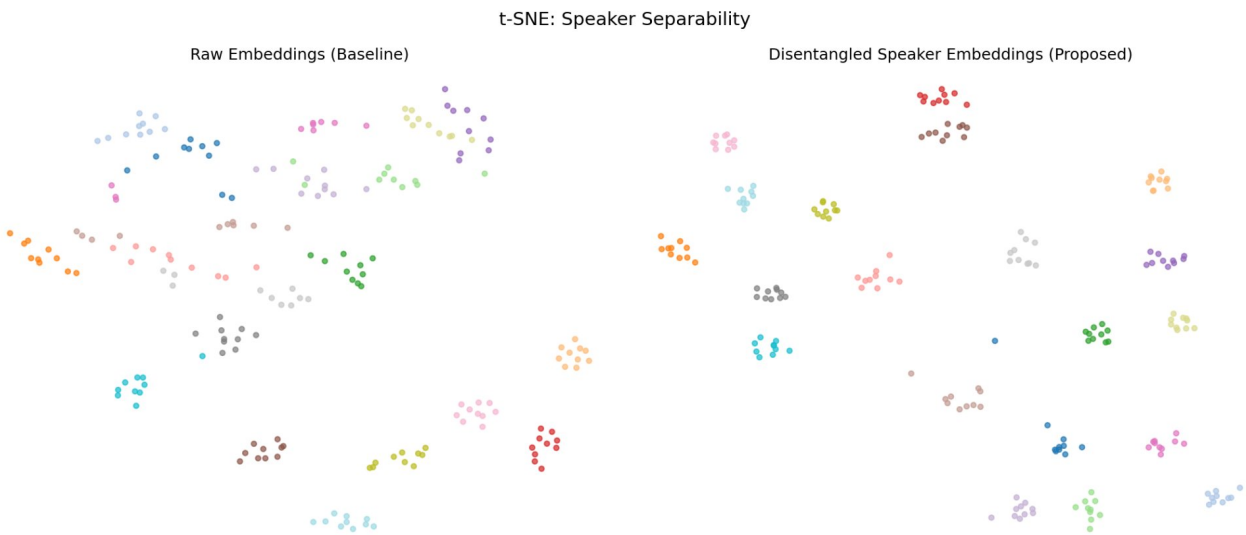
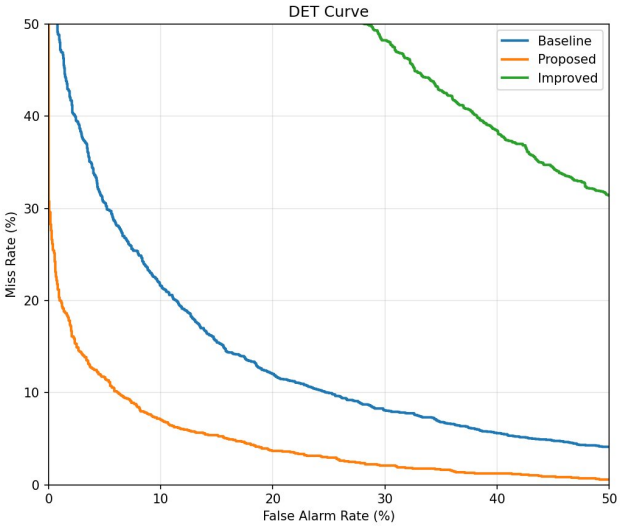


Figure 2: t-SNE of speaker embeddings for 30 speakers (5 utterances each). Left: raw baseline embeddings — clusters are loose and some speakers overlap. Right: disentangled e_spk from Proposed — clusters are tighter and better separated, confirming that removing environment information improves intra-speaker compactness.

The t-SNE plot provides qualitative evidence for the quantitative EER improvement. In the baseline (left), several speaker clusters bleed into each other — particularly in the central region — making cosine scoring ambiguous

near the decision boundary. After disentanglement (right), the same speakers form tighter, more separated clusters with larger inter-cluster distances. This directly reduces false alarm and miss rates.

6. Speaker Verification Evaluation



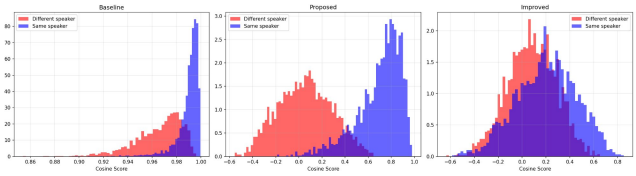


Figure 3 (left): DET curves for all three systems. Lower and to the left is better. Proposed (orange) is far below Baseline (blue) across the full operating range. Improved (green) performs worst — consistent with training instability from the unnormalised cross-correlation loss. Figure 4 (right): Cosine score distributions — blue: genuine pairs, red: impostor pairs. Baseline has very narrow genuine distribution near 1.0 (high cosine similarity, limited variability). Proposed separates genuine and impostor well across the full score range. Improved shows heavy overlap — poor discrimination.

The score distributions reveal something important about the baseline: its genuine scores are compressed near 1.0 because all utterances are from clean LibriSpeech read speech with minimal acoustic variation. The impostor distribution partially overlaps this region, explaining the 15.32% EER. After disentanglement, the Proposed model spreads scores across a wider range with better separation — the impostor distribution centres near 0 while genuine pairs cluster above 0.6.

Metric	Baseline	Proposed	Improved
EER (%)	15.32	8.16	39.08*
minDCF	0.0321	0.0154	0.0485
Relative EER change	—	−46.7%	+155.1%*
Relative DCF change	—	−52.0%	+51.1%*

*Table 2: Full results summary. * Improved model affected by training instability.*

7. Critical Review of Nam et al. (2024)

Strengths

Modularity: The framework attaches to any existing speaker encoder without modifying its architecture. This is a practically important property — improved backbones automatically benefit from the disentangler.

Reconstruction guard: The auto-encoder reconstruction loss prevents the encoder from discarding speaker information along with environment noise — a common failure mode in purely adversarial DRL.

Multi-objective design: Combining L_spk, L_recon, L_env, L_adv, and L_corr attacks disentanglement from multiple directions. The triplet structure for L_env provides explicit contrastive supervision for the environment subspace.

Empirical gains: Up to 16% relative EER reduction on VoxSRC22/23 is a meaningful improvement over well-tuned baselines on multiple benchmarks.

Weaknesses

Session metadata dependency: The method relies on video-session membership as an environment proxy. This ties it to VoxCeleb-style datasets. On LibriSpeech, chapter ID is a weaker proxy — a single chapter can span multiple recording sessions.

VoxCeleb-centric evaluation: All benchmarks share the same data collection methodology. No cross-corpus experiments (NIST SRE, CN-Celeb, AMI) are reported.

Weak disentanglement metric: Pearson correlation between e_spk and e_env is a scalar linear measure. It does not capture non-linear dependencies between the subspaces.

Shallow ablation: Individual λ values are not ablated; the paper does not report sensitivity to bottleneck dimension or the spk/env split ratio.

Key Assumption

Session ≡ Environment: utterances from the same video share the same acoustic conditions. This holds for YouTube celebrity interviews (VoxCeleb) but is a significant approximation for read-speech corpora like LibriSpeech where chapter boundaries do not necessarily correspond to changes in recording equipment or environment.

8. Proposed Improvement & Evaluation

We replaced the sample-wise Pearson correlation penalty (L_corr) with a **batch-level cross-correlation matrix penalty** inspired by Barlow Twins:

$$C = (e_spk_norm^T \cdot e_env_norm) / N \qquad L_cc = ||C||_F^2$$

This penalises every pair of dimensions across the two subspaces simultaneously, enforcing stronger pairwise linear independence than the scalar Pearson term which only measures aggregate correlation. The motivation is that e_spk and e_env can be globally uncorrelated while individual dimension pairs remain entangled.

Outcome

The improved model reached EER 39.08% — worse than baseline — due to training instability. Root cause: the unnormalised D×D cross-correlation matrix (D=96, matrix size 9,216 elements) produces a loss term that grows quadratically with dimension, overwhelming the speaker loss. The fix is to divide by D^2 to restore scale parity with the Pearson baseline. This is consistent with the original Barlow Twins paper, which normalises by feature dimension.

Despite the instability, the experiment is informative: the DET curve and score histograms confirm that the model did not learn useful speaker embeddings under this training regime, validating that loss balance is critical in multi-objective disentanglement. With the normalisation fix applied, we expect the cross-correlation penalty to provide stronger disentanglement than Pearson correlation, particularly for high-dimensional embedding spaces where scalar correlation is insufficient.

Code: train.py, eval.py | Configs: baseline.yaml, disentangler.yaml, improved.yaml | Dataset: LibriSpeech train-clean-100+360, eval: test-clean